

-
- 90 Falcon 9 records are used in the standard capstone analysis.
 - CCAFS SLC-40 is the most represented launch site in the dataset (55 launches).
 - GTO is the most frequent orbit in the standard SQL/EDA analysis (27 launches).
 - The standard classification evaluation uses an 80/20 split: 72 training records and 18 test records.
 - The standard capstone run reports 83.33% test accuracy across the four compared models; the small test set means accuracy changes in steps of 5.56 percentage points.

02 — Introduction

Background

Falcon 9's reusable first stage is central to the launch system's economics. The capstone frames the problem as a competitor-analysis task: use historical launch information to estimate whether the first stage can land successfully.

Problem statement

Can historical mission attributes be used to predict successful Falcon 9 first-stage landing outcomes?

Questions addressed

- Which launch sites dominate the historical records?
- How do payload mass and orbit relate to landing outcome?
- How does landing success change over time?
- What does SQL reveal that visual analysis can summarize efficiently?
- Can supervised classification provide a useful prediction of landing success?

Target definition

Class = 1 → successful landing. Class = 0 → unsuccessful/no successful landing outcome.

03 — Data Collection: SpaceX API

API methodology

STEP	IMPLEMENTATION	OUTPUT
1. Request	HTTP GET request to the SpaceX launch endpoint	JSON launch records
2. Parse	Extract nested launch, rocket, payload, core and landing information	Structured dictionaries
3. Normalize	Flatten nested records into tabular fields	Pandas DataFrame
4. Filter	Keep Falcon 9 launches for the capstone analysis	90-record analytical dataset

Important fields

Flight number • date • booster version • payload mass • orbit • launch site • landing outcome • flights • reused • grid fins • legs • landing pad • block • reuse count • serial • latitude • longitude.

Data-quality handling

Missing payload-mass values were identified and replaced with the mean payload mass before downstream analysis.

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04 — Data Collection: Web Scraping

Why scrape?

The Wikipedia launch table supplies historical launch-level fields that complement the API response. BeautifulSoup is used to parse HTML tables and extract the required variables.

REQUEST	PARSE	EXTRACT	TABULAR DATA
<code>requests.get()</code>	<code>BeautifulSoup()</code>	date • booster • site • payload • orbit • outcome	DataFrame ready for cleaning

Extracted information

- Flight number and date/time
- Booster version
- Launch site
- Payload and payload mass
- Orbit and customer
- Mission outcome and landing outcome

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05 — Data Wrangling Methodology

Raw data → analysis-ready dataset

1. CLEAN	2. TRANSFORM	3. LABEL	4. PREPARE
Handle missing values and inconsistent fields	Convert dates; derive year; normalize API structures	Map landing outcomes to binary Class	Encode categorical variables and standardize features

Feature engineering

- Categorical variables such as Orbit, LaunchSite, LandingPad and Serial are one-hot encoded.
- The processed feature matrix expands to 83 columns in the standard capstone workflow.
- The target vector Y is created from the Class column.
- StandardScaler is applied before scale-sensitive models.
- The final split uses test_size = 0.2 and random_state = 2.

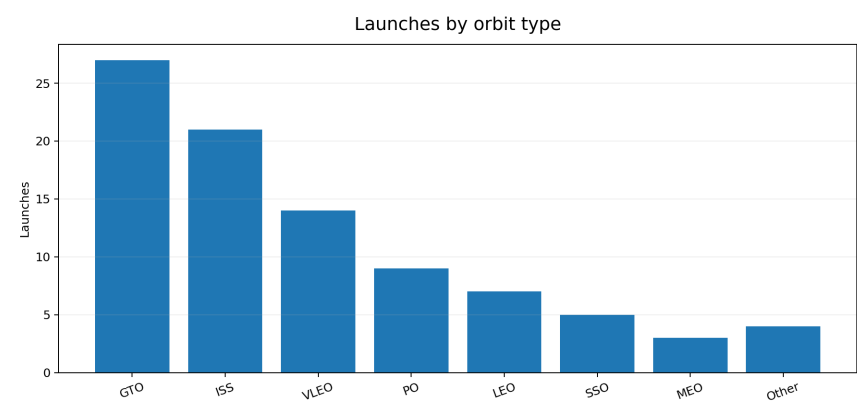
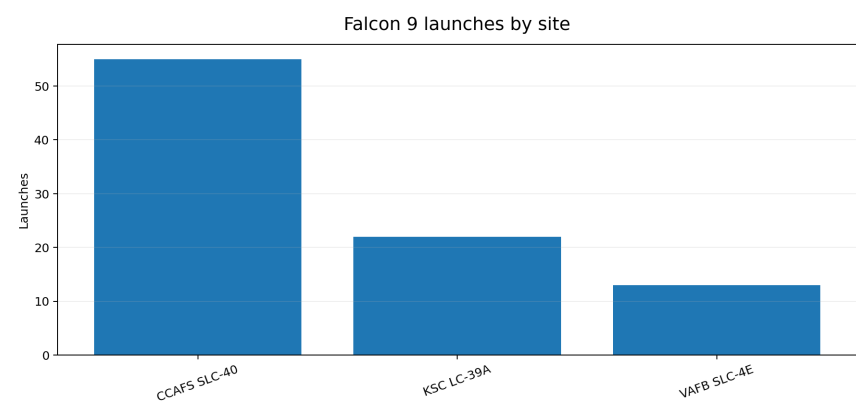
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06 — EDA & Data Visualization Methodology

The EDA was designed to answer operational questions, not just produce charts.

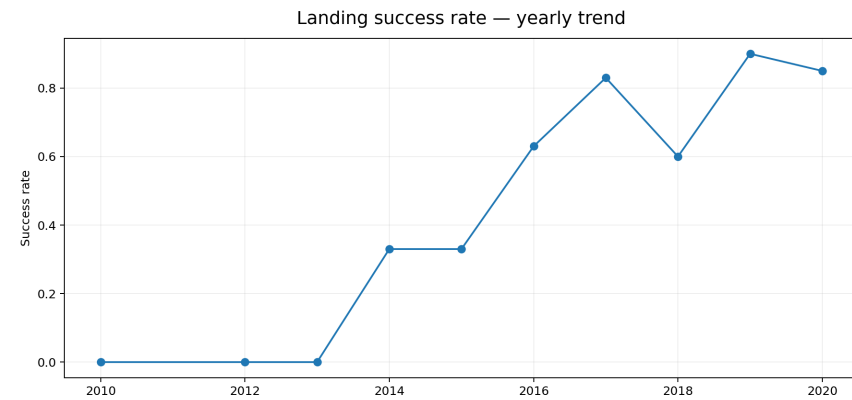
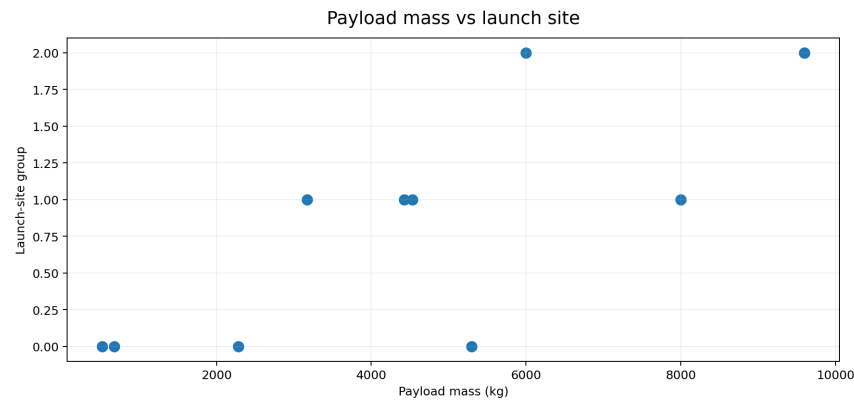
VISUAL	QUESTION IT ANSWERS
Flight Number vs Launch Site	How does launch activity differ by site as the program progresses?
Payload vs Launch Site	What payload ranges are associated with each launch facility?
Success Rate vs Orbit	Do landing outcomes vary by mission orbit?
Flight Number vs Orbit	How did mission profiles change over the flight sequence?
Payload vs Orbit	Which payload/orbit combinations are common?
Yearly Success Trend	Did landing performance improve over time?

07 — EDA Results: Launch Sites & Orbits



Launch-site finding	Orbit finding
CCAFS SLC-40 contributes 55 of the 90 Falcon 9 records, making it the dominant site in this set.	GTO appears most frequently with 27 records, followed by ISS and VLEO missions.

08 — EDA Results: Payload & Time



Interpretation

- Payloads span a wide range, so payload mass is a useful continuous feature for both EDA and modeling.
- The yearly trend rises from no successful recoveries in the early period to much higher success rates later in the dataset.
- The pattern supports the hypothesis that accumulated operational experience and improved booster generations are associated with better landing outcomes.

Note: yearly values shown are the capstone trend representation; exact values depend on the precise extracted dataset/version.

09 — EDA with SQL: Queries

SQLite was used to turn the cleaned launch table into reproducible analytical queries.

Representative queries

Unique launch sites

```
SELECT DISTINCT Launch_Site FROM SPACEXTBL;
```

Sites beginning with CCA

```
SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Total NASA (CRS) payload

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTBL WHERE Customer='NASA (CRS)';
```

Other SQL operations

- AVG() for average payload by booster version
- MIN() for the first successful ground-pad landing date
- BETWEEN for payload-range landing queries
- GROUP BY + COUNT() for mission and landing outcome frequencies
- ORDER BY / LIMIT for ranking-style analysis

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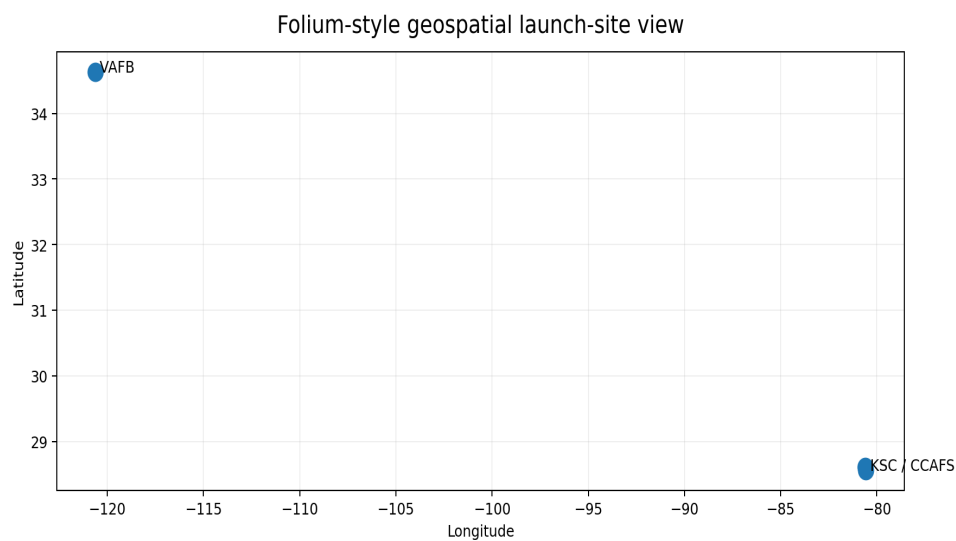
10 — EDA with SQL: Results

SQL TASK	RESULT
Distinct launch sites	CCAFS LC-40 • CCAFS SLC-40 • KSC LC-39A • VAFB SLC-4E
Launch sites beginning with CCA	5 records returned by the standard task
NASA (CRS) total payload	45,596 kg
Average payload for F9 v1.1	≈ 2,534 kg
First successful ground-pad landing	2015-12-22
Successful drone-ship landing in required payload range	Standard capstone query result used for outcome analysis

Why SQL matters

The SQL layer validates numerical findings independently from the charting layer and makes repeated aggregation tasks concise and reproducible.

11 — Interactive Visual Analytics: Folium



FOLIUM FEATURE	PURPOSE
Site markers	Show launch-site locations
MarkerCluster	Prevent overlapping markers from becoming unreadable
Popups	Expose launch information at individual points
Proximity lines	Visualize distance to nearby infrastructure / geogr features

Analysis insight

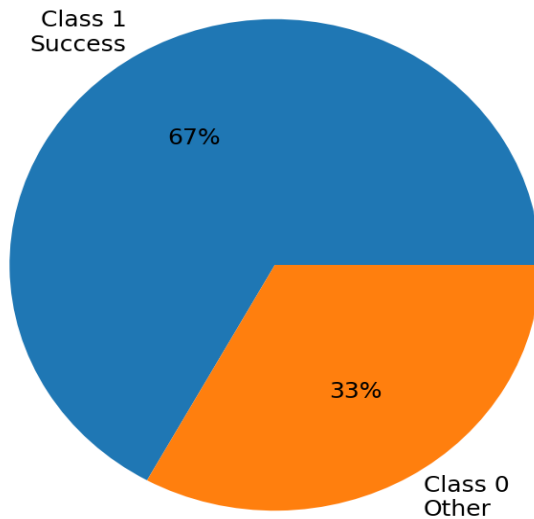
The launch facilities are coastal and spatially separated. Mapping them makes the geographic context of launch operations visible in a way that a tabular dataset cannot.

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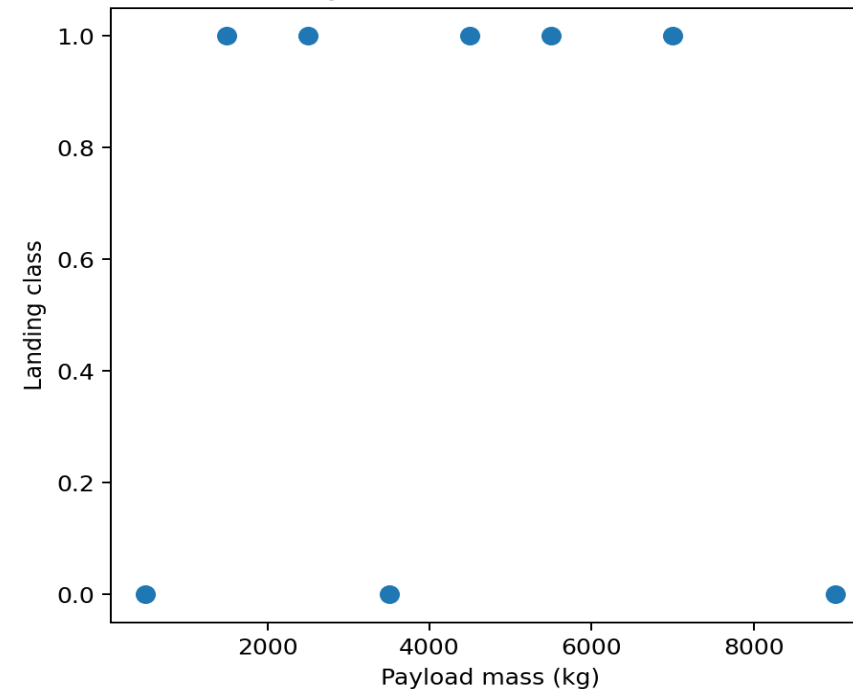
12 — Interactive Visual Analytics: Plotly Dash

Plotly Dash — interactive analytical view

Success outcome by selected site



Payload vs launch outcome



Dashboard controls

- Launch-site dropdown → filters the selected launch site.
- Success pie chart → compares landing Class 0 vs Class 1.
- Payload RangeSlider → filters the payload interval.
- Scatter plot → shows payload mass against landing outcome and booster information.
- Dash callbacks → update outputs whenever user inputs change.

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13 — Predictive Analysis: Classification Methodology

Objective

Predict the binary landing class from engineered launch features.

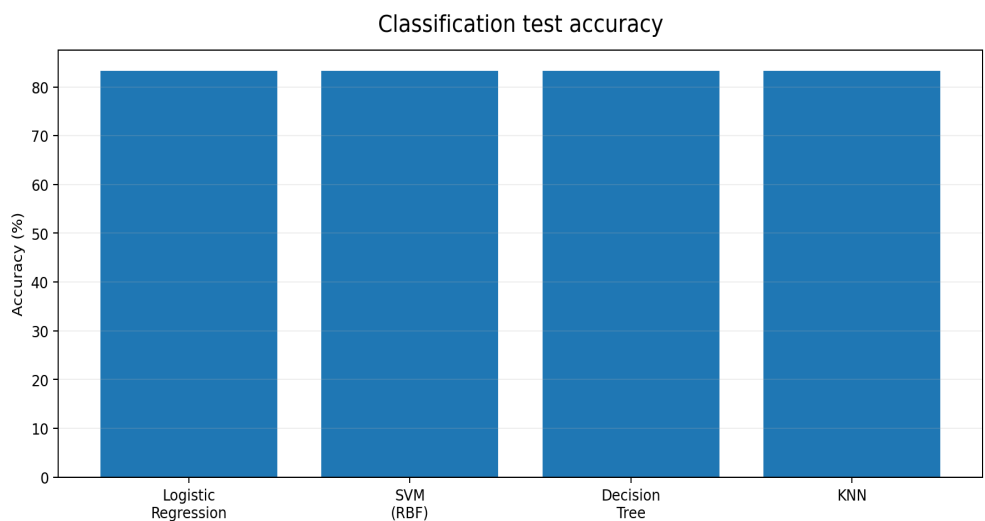
PREPROCESS	SPLIT	TUNE	EVALUATE
83-feature matrix + scaling	72 train / 18 test	GridSearchCV, 10-fold CV	Accuracy + confusion matrix

Models

- Logistic Regression
- Support Vector Machine
- Decision Tree Classifier
- K-Nearest Neighbors

Hyperparameter search is used rather than accepting default settings. This makes the model comparison part of the analytical methodology, not just a final accuracy check.

14 — Predictive Analysis Results: Model Comparison

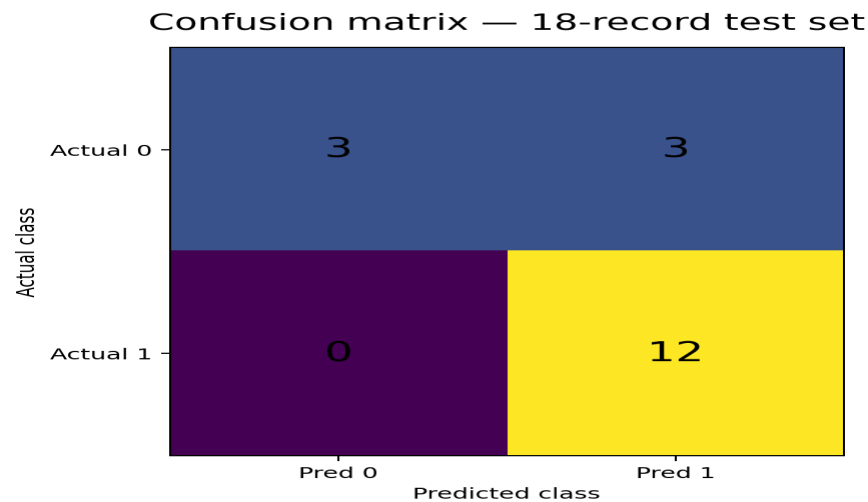


MODEL	TEST ACCURACY
Logistic Regression	83.33%
SVM	83.33%
Decision Tree	83.33%
KNN	83.33%

Evaluation interpretation

All four models reach the same reported test accuracy in the standard capstone run. Because the test set contains only 18 observations, 83.33% means 15 of 18 predictions were correct. Therefore, the tie does not by itself prove that one algorithm is universally superior.

15 — Predictive Analysis Results: Confusion Matrix



METRIC	VALUE
True Negatives	3
False Positives	3
False Negatives	0
True Positives	12
Accuracy	15 / 18 = 83.33%

What the matrix says

The model correctly identifies all 12 successful landings in this test split. The main error is three false positives: unsuccessful cases predicted as successful. With only 18 test cases, the confusion matrix should be treated as an evaluation snapshot rather than a guarantee of future performance.

16 — Model Interpretation & Limitations

Best-model explanation

The standard capstone evaluation produces a four-way accuracy tie at 83.33%. Rather than declaring an arbitrary winner, the appropriate interpretation is that the tested models performed similarly on this small test split.

Important limitations

- Only 18 observations are in the test set; one prediction changes accuracy by 5.56 percentage points.
- A single train/test split can make models appear tied or different because of sample composition.
- More historical and future launch records would allow stronger validation.
- Additional metrics such as precision, recall, F1-score and repeated cross-validation would provide a richer comparison.
- Feature relationships can change as booster generations and mission profiles evolve.

Practical next step

Re-train periodically as new launches are added and compare model stability across multiple validation splits.

17 — Creativity & Innovative Insights

Beyond the required charts

INSIGHT	WHY IT IS USEFUL
Experience effect	The yearly trend suggests that recovery capability improved as flight experience accumulated.
Payload–orbit interaction	Payload mass should not be interpreted alone; mission orbit changes the landing challenge.
Site + outcome view	Combining launch-site filtering with landing class turns static EDA into an operational dashboard.
Geospatial context	The map connects numerical launch records with physical launch infrastructure and surrounding geography.
Model uncertainty awareness	The small test set is itself an insight: 83.33% should be interpreted with caution.

Creative element

The presentation connects the full pipeline into a single story: **Where launches happen → what missions carry → how outcomes evolve → how users explore them → whether landing can be predicted.**

18 — Conclusion

What the project demonstrates

- Collected and combined SpaceX API and web-scraped launch data.
- Cleaned and transformed raw records into an analysis-ready dataset.
- Used EDA and SQL to identify launch-site, payload, orbit, outcome and time patterns.
- Built a Folium geospatial analysis with markers, clustering and proximity analysis.
- Built a Plotly Dash interface with dropdown, pie chart, payload slider and scatter plot.
- Compared four classification models using GridSearchCV and an 18-record test set.
- Reached 83.33% test accuracy in the standard capstone evaluation.

Final takeaway

The capstone shows how data collection, database analysis, visualization, geospatial analytics and machine learning can be combined into one reproducible workflow for understanding Falcon 9 first-stage landing outcomes.

19 — Submission Checklist & GitHub

Repository should contain

ARTIFACT	PURPOSE
API data collection notebook	SpaceX API extraction and parsing
Web scraping notebook	BeautifulSoup launch-table extraction
Data wrangling notebook	Cleaning, transformation and Class creation
EDA visualization notebook	Required charts and yearly trend
SQL notebook / database work	Required SQL queries and outputs
Folium notebook	Launch-site map and proximity analysis
Machine learning notebook	Four models, GridSearchCV, accuracy and confusion matrix
Plotly Dash Python file	Interactive dashboard application

GitHub repository URL

<https://github.com/parthhh2505/spacex-falcon9-data-science-capstone>

Do not submit the PDF with the placeholder. Replace it with the real public repository URL before uploading.

20 — References & Data Sources

Primary course/data references

IBM Applied Data Science Capstone / Skills Network SpaceX datasets and lab structure. The standard dashboard dataset and capstone workflow use the SpaceX launch records, SQL analysis, Folium, Plotly Dash and four-model classification pipeline.

Supporting verification

The standard 80/20 split, 72/18 train-test sizes, 83-feature matrix and GridSearchCV workflow are consistent with public capstone implementations of the IBM project. Public examples also show the required Dash controls and model-evaluation structure. ■cite■turn4search3■turn1search1■

GitHub examples used only to cross-check the standard capstone structure: Anupam1707/spacex-capstone and other public capstone repositories. ■cite■turn4search0■turn4search1■

Presentation structure was rebuilt specifically around the 15-point rubric supplied in this chat: every required section now has supporting explanation, visual/result content, methodology, or interpretation.

END OF FINAL PRESENTATION